Algebra I: Class Notes

Wilson Pan

March 20, 2026

Contents

1 Lecture 1: Group Actions & The Orbit-Stabilizer Theorem (Jan 12) 2
1.1 Permutations and Automorphisms . . . 2
1.2 Group Actions . . . 2
1.3 Orbits and Stabilizers . . . 2

2 Lecture 2: Cauchy's Theorem & Conjugation (Jan 14) 4
2.1 Cauchy's Theorem . . . 4
2.2 Conjugation . . . 4

3 Lecture 3: Sylow Preliminaries & Isomorphism Theorems (Jan 16) 6
3.1 Product of Subgroups . . . 6
3.2 Sylow Definitions . . . 6

4 Lecture 4: The Sylow Theorems (Jan 21) 7

5 Lecture 5: Normal Series & Solvability (Jan 23) 10
5.1 Normal and Subnormal Series . . . 10
5.2 Solvable Groups . . . 10

6 Lecture 6: Solvable and Derived Series (Jan 26) 11

7 Lecture 7: Nilpotent Groups and Free Groups (Jan 28) 13
7.1 Nilpotent Groups . . . 13
7.2 Free Groups . . . 14

8 Lecture 8: Free Groups 15

9 Lecture 9: Generators and Relations (Feb 2) 17

10 Lecture 10: Presentation of Groups (Feb 4) 19

11 Lecture 11: Category Theory (Feb 6) 20
11.1 Categories . . . 20

12 Lecture 12: Ring Theory (Feb 8) 22

13 Lecture 13: Ideals and Quotient Rings (Feb 10) 23

14 Lecture 14: Zorn's Lemma and Modules (Feb 13) 25
14.1 Modules 25

15 Lecture 15: Modules (Feb 16) 26

16 Lecture 16: Integral Domains 28

17 Lecture 17: PID (Feb 19) 30

18 Lecture 18: Unique Factorization Domains (Feb 23) 32

19 Lecture 19: (Feb 25) 34

20 Lecture 20: (Mar 2) 36

21 March 11 39

22 March 11 41
22.1 Goal and Preliminaries 41
22.2 Construction of the Tensor Product 41
22.3 Tensor Product with the Base Ring 43

23 March 16 44

24 March 18 46

25 March 20 48
25.1 Adjoint Functors 48

1 Lecture 1: Group Actions & The Orbit-Stabilizer Theorem (Jan 12)

1.1 Permutations and Automorphisms

Definition 1.1 (Symmetric Group)**.** *Let X be a set. The set of all permutations (bijections) of X is denoted by Sym_X (or sometimes Σ_X). Under function composition, Sym_X forms a group.*

Definition 1.2 (Automorphism Group)**.** *Let $(G, \cdot)$ be a group. An automorphism of G is a bijection $\phi : G \to G$ that is also a homomorphism, i.e.,*

$$\phi(g_1 g_2) = \phi(g_1)\phi(g_2) \quad \forall g_1, g_2 \in G.$$

The set of all such automorphisms is denoted by $\mathrm{Aut}(G)$. It forms a group under composition.

1.2 Group Actions

Definition 1.3 (Group Action)**.** *Let G be a group and X be a set. An **action** of G on X is a homomorphism $\phi : G \to \mathrm{Sym}_X$.*
We typically write the action as $g \cdot x := \phi(g)(x)$. This notation satisfies two axioms (equivalent to the homomorphism property):

1. ***Identity:*** *$1 \cdot x = x$ for all $x \in X$.*
2. ***Compatibility:*** *$g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x$ for all $g_1, g_2 \in G, x \in X$.*

Theorem 1.4 (Equivalence Relation on X)**.** *Let G act on X. Define a relation $\sim$ on X by:*

$$x_1 \sim x_2 \iff \exists g \in G \text{ such that } g \cdot x_1 = x_2.$$

Then $\sim$ is an equivalence relation.

Proof. We check the properties:

1. **Reflexive:** $1 \cdot x = x \implies x \sim x$.
2. **Symmetric:** If $g \cdot x_1 = x_2$, acting by g^{-1} gives $g^{-1} \cdot (g \cdot x_1) = g^{-1} \cdot x_2 \implies (g^{-1}g) \cdot x_1 = g^{-1} \cdot x_2 \implies x_1 = g^{-1} \cdot x_2$. Thus $x_2 \sim x_1$.
3. **Transitive:** If $g \cdot x_1 = x_2$ and $h \cdot x_2 = x_3$, then $h \cdot (g \cdot x_1) = x_3 \implies (hg) \cdot x_1 = x_3$. Thus $x_1 \sim x_3$.

□

1.3 Orbits and Stabilizers

The equivalence classes under the relation $\sim$ partition the set X. These classes are called **orbits**.

Definition 1.5 (Orbit)**.** *For $x \in X$, the orbit of x is the set of all places x can be moved to by G:*

$$\mathcal{O}_x = Orb(x) = \{g \cdot x : g \in G\}.$$

Definition 1.6 (Stabilizer)**.** *For $x \in X$, the stabilizer of x is the set of elements in G that fix x:*

$$G_x = \mathrm{Stab}(x) = \{g \in G : g \cdot x = x\}.$$

Note that G_x is a subgroup of G (denoted $G_x \leq G$).

Theorem 1.7 (**Orbit-Stabilizer Theorem**)**.** *Let G act on X. For any $x \in X$, there is a bijection between the orbit $\mathcal{O}_x$ and the set of left cosets G/G_x. Consequently:*

$$|\mathcal{O}_x| = [G : G_x].$$

If G is finite, $|G| = |\mathcal{O}_x| \cdot |G_x|$.

Proof. Define a map $\psi : G/G_x \to \mathcal{O}_x$ by $\psi(gG_x) = g \cdot x$.

1. **Well-defined:** Suppose $g_1G_x = g_2G_x$. Then $g_1^{-1}g_2 \in G_x$, so $(g_1^{-1}g_2) \cdot x = x$, implying $g_2 \cdot x = g_1 \cdot x$.
2. **Injectivity:** If $\psi(g_1G_x) = \psi(g_2G_x)$, then $g_1 \cdot x = g_2 \cdot x$. Multiplying by g_1^{-1}, we get $x = g_1^{-1}g_2 \cdot x$, so $g_1^{-1}g_2 \in G_x$, implying $g_1G_x = g_2G_x$.
3. **Surjectivity:** By definition, any $y \in \mathcal{O}_x$ is of the form $g \cdot x$ for some g, which is exactly $\psi(gG_x)$.

Thus, ψ is a bijection. $\square$

2 Lecture 2: Cauchy's Theorem & Conjugation (Jan 14)

2.1 Cauchy's Theorem

Lemma 2.1. *If X is a finite set and G is a p-group (a group of order p^k) acting on X, then:*

$$|X| \equiv |X^G| \pmod{p},$$

where $X^G = \{x \in X : g \cdot x = x, \forall g \in G\}$ is the set of fixed points.

Theorem 2.2 (Cauchy's Theorem)**.** *If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p.*

Proof (McKay's Proof). Let X be the set of p-tuples of elements of G whose product is the identity:

$$X = \{(g_1, \ldots, g_p) \in G^p : g_1 g_2 \cdots g_p = 1\}.$$

Notice that g_p is uniquely determined by the first $p-1$ elements ($g_p = (g_1 \cdots g_{p-1})^{-1}$), so $|X| = |G|^{p-1}$. Since $p \mid |G|$, we have $p \mid |X|$.
Let $\mathbb{Z}_p$ (cyclic group of order p) act on X by cyclic shift:

$$\sigma \cdot (g_1, g_2, \ldots, g_p) = (g_2, g_3, \ldots, g_p, g_1).$$

This is a valid action because if $g_1 \cdots g_p = 1$, then $g_1(g_2 \cdots g_p) = 1 \implies g_2 \cdots g_p = g_1^{-1}$, so $(g_2 \cdots g_p)g_1 = g_1^{-1} g_1 = 1$.
By the Lemma, $|X| \equiv |X^{\mathbb{Z}_p}| \pmod{p}$. Fixed points are tuples $(a, a, \ldots, a)$ such that $a^p = 1$. Since $(1, \ldots, 1) \in X^{\mathbb{Z}_p}$, the set of fixed points is non-empty. Since p divides $|X|$ and p divides the congruence $|X| - |X^{\mathbb{Z}_p}|$, p must divide $|X^{\mathbb{Z}_p}|$. Therefore, there are at least p fixed points. There must exist some $a \neq 1$ such that $a^p = 1$. □

2.2 Conjugation

Conjugation is a specific action of G on itself.

Definition 2.3. *For $g, h \in G$, the **conjugate** of g by h is $g^h := hgh^{-1}$. The map $\phi_h : G \to G$ defined by $g \mapsto hgh^{-1}$ is an automorphism (an inner automorphism).*

Definition 2.4 (Classes and Centralizers)**.** *Let G act on itself by conjugation ($h \cdot g = hgh^{-1}$).*

- *The orbit of g is the **Conjugacy Class** of g: $Cl(g) = \{hgh^{-1} : h \in G\}$.*
- *The stabilizer of g is the **Centralizer** of g: $C_G(g) = \{h \in G : hg = gh\}$.*

By Orbit-Stabilizer: $|Cl(g)| = [G : C_G(g)]$.

Definition 2.5 (Center of G)**.** *The* ***Center****, $Z(G)$, is the set of elements that commute with everything:*

$$Z(G) = \{z \in G : zg = gz, \forall g \in G\} = \bigcap_{g \in G} C_G(g).$$

$Z(G)$ is the kernel of the conjugation homomorphism $G \to \mathrm{Aut}(G)$. Thus $Z(G) \trianglelefteq G$.

Definition 2.6 (Normalizer)**.** *Let $H \leq G$. The* ***Normalizer*** *of H in G is:*

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

$N_G(H)$ is the largest subgroup of G in which H is normal.

3 Lecture 3: Sylow Preliminaries & Isomorphism Theorems (Jan 16)

3.1 Product of Subgroups

Definition 3.1. *Let $K, N \leq G$. Define $KN = \{kn : k \in K, n \in N\}$.*

- *In general, KN is not a subgroup.*
- *If $N \trianglelefteq G$ (or just $N \subseteq N_G(K)$), then KN is a subgroup.*
- *Size formula: $|KN| = \frac{|K||N|}{|K \cap N|}$.*

Theorem 3.2 (Second Isomorphism Theorem)**.** *Let $K \leq G$ and $N \trianglelefteq G$. Then $K \cap N \trianglelefteq K$, and*

$$\frac{KN}{N} \cong \frac{K}{K \cap N}.$$

Proof. Consider the natural projection $\pi : G \to G/N$. Restrict it to K, i.e., $\phi = \pi \mathrm{res}_K : K \to G/N$. The image of ϕ is $\{kN : k \in K\} = KN/N$. The kernel of ϕ is $\{k \in K : kN = N\} = \{k \in K : k \in N\} = K \cap N$. By the First Isomorphism Theorem, $K/\ker(\phi) \cong \mathrm{Im}(\phi)$. □

3.2 Sylow Definitions

Definition 3.3 (p-group)**.** *Let p be a prime. A group G is a p-group if every element has order a power of p. For finite groups, this is equivalent to $|G| = p^k$.*

Definition 3.4 (Sylow p-subgroup)**.** *Let $|G| = p^n m$ where $p \nmid m$. A subgroup $P \leq G$ is called a **Sylow p-subgroup** if $|P| = p^n$.*

$$Syl_p(G) = \{P \leq G : P \text{ is a Sylow } p\text{-subgroup}\}.$$

4 Lecture 4: The Sylow Theorems (Jan 21)

Theorem 4.1 (**The Sylow Theorems**). *Let G be a finite group of order $p^n m$ where $p \nmid m$.*

1. ***Existence:*** *$Syl_p(G) \neq \emptyset$. (There exists a subgroup of order p^n).*
2. ***Conjugacy:*** *Any two Sylow p-subgroups are conjugate in G. That is, if $P, Q \in Syl_p(G)$, $\exists g \in G$ such that $gPg^{-1} = Q$.*
3. ***Number:*** *Let $n_p = |Syl_p(G)|$. Then:*
   - $n_p \equiv 1 \pmod{p}$.
   - $n_p \mid m$ *(equivalently $n_p \mid |G|$).*

Proof. Let p be a prime such that $p \,\Big|\, |G|$. Define the set of all p-subgroups:

$$\Sigma = \{H : H \leq G, |H| = p^n \text{ for some } n > 0\}.$$

Define Ω as the set of maximal elements in Σ under inclusion:

$$\Omega = \{H : H \in \Sigma, \text{ there is no } K \in \Sigma \text{ such that } H \subsetneq K\}.$$

We let G act on the set of subgroups $\{H : H \leq G\}$ by conjugation. Since conjugation is an isomorphism ($H^g \simeq H$), it preserves the order of subgroups. Thus:

$$H \in \Sigma \iff H^g \in \Sigma \quad \text{and} \quad H \in \Omega \iff H^g \in \Omega.$$

Therefore, G acts on Ω by $g \cdot H = H^g$.

Claim 1: Let $H \in \Omega$. Consider the action of H on Ω by conjugation. Then H is the *unique* fixed point of this action.

Proof of Claim:

- **Existence:** It is trivial to see that H is a fixed point, as $H^h = hHh^{-1} = H$ for any $h \in H$.
- **Uniqueness:** Let $K \in \Omega$ be a fixed point of the action of H.

  $$K \text{ is fixed by } H \iff K^h = K \quad \forall h \in H \iff H \leq N_G(K).$$

  Since $K \trianglelefteq N_G(K)$, and $H \leq N_G(K)$, the product HK is a subgroup of $N_G(K)$, and thus $HK \leq G$. Using our counting lemma:

  $$|HK| = \frac{|H||K|}{|H \cap K|}.$$

  Since $H, K \in \Omega$, their orders are powers of p. Thus $|HK|$ is a power of p, meaning $HK \in \Sigma$.

  We have $H \leq HK$. Since H is maximal in Σ (by definition of Ω) and $HK \in \Sigma$, it must be that $H = HK$. Similarly, $K \leq HK$ implies $K = HK$. Therefore, $H = K$.

Claim 2: Size of Ω Modulo p

Statement: $|\Omega| \equiv 1 \pmod{p}$.

Proof of Claim: Let $H \in \Omega$. We decompose Ω into disjoint orbits under the action of H.

$$\Omega = \{H\} \cup \bigcup_{K \neq H} \mathrm{Orb}_H(K).$$

By Claim 1, H is the only fixed point (an orbit of size 1). For any $K \in \Omega$ with $K \neq H$, the stabilizer of K in H is a proper subgroup, so by the Orbit-Stabilizer theorem for p-groups, $|\mathrm{Orb}_H(K)|$ is divisible by p.

$$|\Omega| = 1 + \sum(\text{multiples of } p) \implies |\Omega| \equiv 1 \pmod{p}.$$

Claim 3: Transitivity (Conjugacy of Elements in Ω)

Statement: Any two elements of Ω are conjugate.
Proof of Claim: Suppose for contradiction that $H, K \in \Omega$ are not conjugates. Let $\theta_1 = \mathrm{Orb}_G(H)$ and $\theta_2 = \mathrm{Orb}_G(K)$. Since they are distinct orbits, $\theta_1 \cap \theta_2 = \emptyset$.

Consider the action of H on these sets:

- On θ_1: $H \in \theta_1$. By Claim 1, H is the unique fixed point of H in Ω. Since $\theta_1 \subseteq \Omega$, H is the unique fixed point in θ_1. Thus, $|\theta_1| \equiv 1 \pmod{p}$.
- On θ_2: Since $H \in \theta_1$ and $\theta_1 \cap \theta_2 = \emptyset$, we have $H \notin \theta_2$. Therefore, θ_2 contains *no* fixed points under the action of H (because the only fixed point in all of Ω is H). Thus, every orbit of H inside θ_2 has size divisible by p. This implies $|\theta_2| \equiv 0 \pmod{p}$.

By symmetry, we can swap the roles of H and K. Running the same argument with K acting on the sets implies $|\theta_2| \equiv 1 \pmod{p}$ and $|\theta_1| \equiv 0 \pmod{p}$.

This results in a contradiction (e.g., $|\theta_1| \equiv 1$ and $|\theta_1| \equiv 0$). Thus, $\theta_1 = \theta_2$, so H and K are conjugate.

Corollary: Since there is only one orbit, $\Omega = \theta_1$. Thus $|\Omega| = |\theta_1| \equiv 1 \pmod{p}$.

Claim 4: Divisibility

Statement: $|\Omega| \,\Big|\, |G|$.
Proof of Claim: Let $H \in \Omega$. Since Ω is exactly the orbit of H under conjugation (from Claim 3), the size of the orbit is the index of the stabilizer:

$$|\Omega| = [G : \mathrm{Stab}_G(H)] = [G : N_G(H)] = \frac{|G|}{|N_G(H)|}.$$

Thus $|\Omega|$ divides $|G|$.

Claim 5: Identification with Sylow Subgroups

Statement: $\Omega = Syl_p(G)$.
Proof of Claim: We know $Syl_p(G) \subseteq \Omega$ because Sylow subgroups are maximal p-subgroups by definition. We must show the reverse: every $H \in \Omega$ is a Sylow p-subgroup. Let $H \in \Omega$. Assume for contradiction that $H \notin Syl_p(G)$. Let $|G| = p^s m$ where $p \nmid m$. Since H is a p-group but not Sylow, $|H| = p^t$ with $t < s$.
Consider the normalizer $N_G(H)$. From Claim 2, we know $|\Omega| = [G : N_G(H)] \equiv 1 \pmod{p}$. Therefore, p does *not* divide $[G : N_G(H)]$. Since $|G| = [G : N_G(H)] \cdot |N_G(H)|$, all factors of p in $|G|$ must reside in $|N_G(H)|$. Thus, $|N_G(H)|$ is divisible by p^s.
Consequently, the index $[N_G(H) : H] = \frac{|N_G(H)|}{|H|} = \frac{(\text{multiple of } p^s)}{p^t}$ is divisible by p (since $s > t$).
By Cauchy's Theorem applied to the quotient group $N_G(H)/H$, there exists a subgroup of order p, say $\bar{K} \leq N_G(H)/H$. By the Correspondence Theorem, there exists a subgroup K such that $H \leq K \leq N_G(H)$ with $|K| = p \cdot |H| = p^{t+1}$.
Since K is a p-group, $K \in \Sigma$. However, $H \subsetneq K$, which contradicts the definition of H as a maximal element in Ω.
Therefore, H must already be of order p^s. Thus $\Omega = Syl_p(G)$. $\square$

5 Lecture 5: Normal Series & Solvability (Jan 23)

5.1 Normal and Subnormal Series

Definition 5.1. *A **subnormal series** of a group G is a chain of subgroups:*

$$\{1\} = H_0 \leq H_1 \leq \cdots \leq H_k = G$$

*such that $H_i \trianglelefteq H_{i+1}$ for all i. The quotient groups H_{i+1}/H_i are called the **factors** of the series.*

Definition 5.2 (Composition Series)**.** *A subnormal series is a **composition series** if all factors H_{i+1}/H_i are **simple** groups (non-trivial groups with no normal subgroups other than $\{1\}$ and themselves).*

Theorem 5.3 (Jordan-Hölder)**.** *Any two composition series of a finite group G are equivalent. That is, they have the same length, and their factors are isomorphic (up to reordering).*

5.2 Solvable Groups

Definition 5.4. *A group G is **solvable** if it has a subnormal series where all factors H_{i+1}/H_i are **abelian**.*

Definition 5.5 (Commutator)**.** *The commutator of g, h is $[g, h] = ghg^{-1}h^{-1}$. The **Derived Subgroup** (or Commutator Subgroup) G' or $[G, G]$ is the subgroup generated by all commutators.*

Proposition 5.6. *G/N is abelian if and only if $[G, G] \leq N$. Thus, $[G, G]$ is the smallest normal subgroup such that the quotient is abelian.*

Remark 5.7 (Derived Series)**.** *Define $G^{(0)} = G$, $G^{(1)} = [G, G]$, and $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. G is solvable if and only if the derived series terminates at $\{1\}$ (i.e., $G^{(n)} = \{1\}$ for some n).*

6 Lecture 6: Solvable and Derived Series (Jan 26)

Lemma 6.1. *Let $\alpha \in Aut(G)$ then $\alpha([g,h]) = [\alpha(g), \alpha(h)]$.*

Lemma 6.2. *For $N \trianglelefteq G$ then G/N is abelian iff $[G,G] \leq N$.*

Theorem 6.3. *If $N \trianglelefteq G$, $g, h \in G$ and $[gN, hN] = [g,h]N$. Additionally, gN and hN commutes iff $[g,h]N = N$ iff $[g,h] \subseteq N$.*

Definition 6.4. *G is solvable iff there is a subnormal series $\langle H_i : 0 \leq i \leq t \rangle$ such that $[H_{i+1}, H_{i+1}] \leq H_i$ for all $0 \leq i \leq t-1$.*

Theorem 6.5. *The following are true*
1. *If G solvable and $K \leq G$ then K is solvable*
2. *If G is solvable and $N \trianglelefteq G$ then G/N is solvable.*
3. *For $N \trianglelefteq G$, G solvable iff both N and G/N are solvable*

Proof. We prove the previous theorems

1. Let $\langle H_i : 0 \leq i \leq t \rangle$ be a series in G such that $[H_{i+1}, H_{i+1}] \leq H_i$ for all $0 \leq i \leq t-1$. We can let $H_i' = K \cap H_i$ for $0 \leq i \leq t$ and we need to verify $\langle H_i' : 0 \leq i \leq t \rangle$ is a solvable series in K.

2. Let $\langle H_i : 0 \leq i \leq t \rangle$ be a series in G such that $[H_{i+1}, H_{i+1}] \leq H_i$ for all $0 \leq i \leq t-1$. Let $H_i' = \phi_N[H_i] = H_iN/N$. Verify it is solvable.

3. The forward direction is trivial by (1) and (2). For the reverse direction, N solvable and G/N solvable $\langle N_i : 0 \leq i \leq s \rangle$ subnormal in N and N_{i+1}/N_i abelian and similar for $\langle H_jN/N : 0 \leq j \leq t \rangle$ subnormal in G/N and $(H_{j+1}N/N)_{j+1}/(H_jN/N)$ abelian. We know $H_{j+1} \trianglelefteq H_j$ and $[H_{j+1}, H_{j+1}] \leq H_j$ for all $0 \leq j \leq t-1$. So $N_0 = 1 \trianglelefteq N_1 \cdots \trianglelefteq N_s = N = H_0 \trianglelefteq H_1 \cdots \trianglelefteq H_t = G$.

□

Definition 6.6. *Given G, the derived series of G is given by $G_0' = G$ and $G_{i+1}' = [G_i', G_i']$ for $i \geq 0$.*

Theorem 6.7. *G is solvable iff there is n such that $G_n' = \{1\}$.*

Proof. For the backwards direction, by construction G_i'/G_{i+1}' is abelian. Let $H_j = G_{n-j}$. For the forward direction, let $\langle H_j : 0 \leq j \leq n \rangle$ be a subnormal series in G, H_{j+1}/H_j abelian. Show by induction that $G_i' \leq H_{n-i}$ for all i. □

Theorem 6.8. *Suppose G abelian and simple, let $a \in G$ and $a \neq 1$ then $G = \langle a \rangle$ and $|a|$ is prime.*

Theorem 6.9. *Let G be simple and let $\langle H_i : 0 \leq i \leq t \rangle$ be subnormal series in G. Then there is $n < t$ such that $H_i = 1$ for $0 \leq i \leq n$ and $H_i = G$ for $n < i \leq t$.*

7 Lecture 7: Nilpotent Groups and Free Groups (Jan 28)

7.1 Nilpotent Groups

Definition 7.1. *For $H, K \leq G$, $[H, K] = \langle [h, k] : h \in H, k \in K \rangle$*

Remark 7.2. *G_i' is characteristic in G for all i, in particular $G_i' \trianglelefteq G$.*

Lemma 7.3. *If $N \trianglelefteq G, N \leq H \leq G$ then*

$$\frac{H}{N} \leq Z\left(\frac{G}{N}\right) \iff [G, H] \leq N.$$

Definition 7.4. *A normal series in G, $\langle H_i : 0 \leq i \leq t \rangle$ is a central series if $\frac{H_{i+1}}{H_i} \leq Z\left(\frac{G}{H_i}\right)$ for all $0 \leq i \leq t-1$. Equivalently, $[G, H_{i+1}] \leq H_i$ for all $0 \leq i \leq t-1$.*

Definition 7.5. *G is nilpotent iff G has a central series.*

Remark 7.6. *Nilpotent groups are solvable.*

Definition 7.7. *Let G be a group*

1. *The descending central series is the sequence of subgroup given by $\gamma_1(G) = G$ and $\gamma_{i+1}(G) = [G, \gamma_i(G)]$ for $i \geq 1$.*
2. *The ascending central series is the sequence of subgroup given by $Z_0(G) = 1$ and*

$$\frac{Z_{n+1}(G)}{Z_n(G)} = Z\left(\frac{G}{Z_n(G)}\right).$$

Remark 7.8. *By induction, $Z_n(G)$ and $\gamma_n(G)$ are characteristic in G. and $Z_n(G) \trianglelefteq G$ and $\gamma_n(G) \trianglelefteq G$ for all $n \geq 0$.*

Remark 7.9. *For any G and $N \trianglelefteq G, [G, N] \leq N$ because for any $g \in G, n \in N$ we have $[g, n] = gng^{-1}n^{-1}$ and $n^g \in N$ since $N \trianglelefteq G$. So $[G, N] \leq N$.*

Theorem 7.10. *G is nilpotent iff there is a $n \geq 0$ such that $Z_n(G) = G$ iff there is $n \geq 0$ such that $\gamma_{n+1}(G) = \{1\}$. Moreover, the least n such that $Z_n(G) = G$ is the least n such that $\gamma_{n+1}(G) = \{1\}$.*

Proof. Take any central series then the decreasing and ascending central series will grow at least as fast as the arbitrary central series. □

7.2 Free Groups

Example 7.11. *Consider $\mathbb{Z} = (\mathbb{Z}, +)$, 1 is a generator $\mathbb{Z} = \langle 1 \rangle$. The universal property of $\mathbb{Z}$, 1. For any group G and any $g \in G$, there is a unique homomorphism $\phi : \mathbb{Z} \to G$ and $\phi(1) = g$.*

Proof. If ϕ exist let $\phi(n) = g^n$ then ϕ is a homomorphism. □

g is free because whatever 1 is mapped to in the homomorphism it won't mess you up compared to a finite group.

Definition 7.12. *F is a free group on 2 element if*

1. *There exist $a, b \in F$, $F = \langle a, b \rangle$*
2. *For all G, all $g, h \in G$ there is a unique homomorphism $\phi : F \to G$ such that $\phi(a) = g$ and $\phi(b) = h$.*

8 Lecture 8: Free Groups

Theorem 8.1. *For any set X, there are a group F and an injective function $i : X \to F$ such that*

1. *For any group G and any function $f : X \to G$ there is a unique homomorphism $\phi : F \to G$ such that $\phi(i(x)) = f(x)$ for all $x \in X$.*

$$\begin{tikzcd} X \arrow[r, hook, "i"] \arrow[dr, "f"'] & F \arrow[d, dashed, "\phi"] \\ & G \end{tikzcd}$$

   The diagram commutes: $\phi \circ i = f$. The homomorphism ϕ is unique.

2. *F is generated by $i[X]$*

Proof. A word (on alphabet X) is a (possibly empty) finite sequence from $X \times \mathbb{Z}$. We will write $x_1^{n_1} x_2^{n_2} \cdots x_k^{n_k}$ for the word $x_1, x_2, \ldots, x_k$ for $\langle (x_1, n_1), (x_2, n_2), \ldots, (x_k, n_k) \rangle$.

Example 8.2. *Let $X = \{x, y, z\}$ then $w = x^0 x^{15} y^{-23} z^{27}$ is a word.*

Let w be the set of words. Given a word, we can possibly perform a reduction step

1. If word contains an entry $\langle x, 0 \rangle$, remove it.
2. If the word contains successive entries $\langle x, n \rangle$ and $\langle x, m \rangle$ then replace them with $\langle x, n + m \rangle$.

Example 8.3. $y^{15} x^3 x^{-2} x^{-1} y^2 z^5 \to y^{15} x^1 x^{-1} y^2 z^5 \to y^{15} x^0 y^2 z^5 \to y^{15} y^2 z^5 \to y^{17} z^5$.

Let R be set of all reduced words, where no more reductions can be done.
For any word w, there is a unique reduced word w' such that w can be transformed into w' by a finite set of reductions.
We will not be bale to show this property using group properties but instead we'll produce a group F with the property that every word is equivalent to a unique reduced word.

Given $x \in X$, I will replace $f_x : F \to F$

1. w does not begin with some x^m then $f_x(w) = xw$
2. w is of the form $x^{-1}v$ then $f_x(w) = v$
3. w is of the form $x^n v$ for $n \neq -1$ then $f_x(w) = x^{n+1} v$

I also define $g_x : R \to R$ by

1. w does not begin with some x^n then $g_x(w) = x^{-1} w$
2. w is of the form $x^1 v$ then $g_x(w) = v$
3. w is of the form $x^n v$ for $n \neq 1$ then $g_x(w) = x^{n-1} v$

We can observe that $f_x \circ g_x = g_x \circ f_x = 1_R$.
For each $x \in X$, $f_x \in \sum_R$ (the group of permutations of R) and $g_x = f_x^{-1}$. Let F be the subgroup of $\sum_R$ generated by $\{f_x : x \in X\}$ or $\{f_{x_1}^{n_1}, ..., f_{x_t}^{n_t} : x_i \in X, n_i \in \mathbb{Z}\}$.

Let $i : X \to F$ with $i(x) = f_x$. For any word $w = x_1^{n_1} x_2^{n_2} \cdots x_t^{n_t}$. Let $f_w = f_{x_1}^{n_1} f_{x_2}^{n_2} \cdots f_{x_t}^{n_t} \in F$.
Key Facts:

1. If w is obtained from $\overline{w}$ by a single reduction step, $f_w = f_{\overline{w}}$.
2. If w is obtained from $\overline{w}$ by any sequence of reduction steps, then $f_w = f_{\overline{w}}$.
3. If $w \in R$ then $f_w(\langle\rangle) = w$
4. For any word w, $f_w(\langle\rangle)$ the unique reduced word to which w can be transformed by reduction step.
5. If w_1, w_2 are reduced words then $v =$ unique reduction of $w_1 w_2$, $f_v = f_{w_1} f_{w_2}$

Let F' be the group whose underlying set is R whose operation is "concatenate and reduce". Then $F \simeq F'$ by $f_v \mapsto v$. To finish, let G be a group, $h : X \to G$ function. We want to find there is a unique homomorphism ϕ with $F' \to G$ and $\phi \circ i = h$.

1. If ϕ exists then

$$\begin{aligned}\phi(x_1^{n_1} x_2^{n_2} \cdots x_t^{n_t}) &= \phi(i(x_1))^{n_1} \phi(i(x_2))^{n_2} \cdots \phi(i(x_t))^{n_t} \\ &= h(x_1)^{n_1} h(x_2)^{n_2} \cdots h(x_t)^{n_t}\end{aligned}$$

2. Verify ϕ is a homomorphism.

$\square$

Example 8.4. *Let $X = \langle x, y\rangle$, let G be any group with 2 generators say $G = \langle a, b\rangle$. Let F be a free group on X. Let $\phi : F \to G$ be unique homomorphism such that $\phi(x) = a$ and $\phi(y) = b$. ϕ is surjective because G is generated by a and b. Thus, $G \simeq \frac{F}{\ker \phi}$.*

9 Lecture 9: Generators and Relations (Feb 2)

Definition 9.1. *let X be a set, let F be a free group on X. Let $A \subseteq F$ and $N =$ least normal subgroup of F containing A.*

Theorem 9.2. *Let $\overline{F} = F/N$ and $j : X \to \overline{F}$ by $j(x) = xN$.*

$$\begin{tikzcd} F \arrow[r, "\phi_N"] & \overline{F} \\ X \arrow[u, "i"] \arrow[ur, "j"] & \end{tikzcd}$$

The following are true:

1. *$\overline{F}$ is generated by $\{j(x) : x \in X\}$. Let $\overline{x} = j(x)$*
2. *If $a = x_1^{n_1} x_2^{n_2} \cdots x_t^{n_t} \in A$ then $a \in N$ and $\overline{x_1}^{n_1} \overline{x_2}^{n_2} \cdots \overline{x_t}^{n_t} = \phi_N(a) = 1_{\overline{F}}$*

"Universal Property of $\overline{F}$": Let G be a group and let $h : X \to G$ such that $a = x_1^{n_1} x_2^{n_2} \cdots x_t^{n_t} \in A$ and $h(x_1)^{n_1} h(x_2)^{n_2} \cdots h(x_t)^{n_t} = 1_G$. There is an unique homomorphism $\mathcal{T} : \overline{F} \to G$ such that

$$\begin{tikzcd} X \arrow[r, "j"] \arrow[dr, "h"'] & \overline{F} \arrow[d, dashed, "\mathcal{T}"] \\ & G \end{tikzcd}$$

So $h = \mathcal{T} \circ j$

Proof. Since $\{\overline{x} : x \in X\}$ generates $\overline{F}$ there is at most one τ. By universal property of F, there is unique homomorphism $\phi : F \to G$ such that

$$\phi(x^1) = h(x). \qquad \text{(For all } x \in X\text{)}$$

Use property of F

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[dr, "h"'] & F \arrow[d, dashed, "\phi"] \\ & G \end{tikzcd}$$

Claim: By hypothesis on h, $A \subseteq \ker(\phi)$.
Let $a = x_1^{t_1} x_2^{t_2} \cdots x_k^{t_k} \in A$ then $\phi(a) = \phi(x_1)^{t_1} \phi(x_2)^{t_2} \cdots \phi(x_k)^{t_k} = h(x_1)^{t_1} h(x_2)^{t_2} \cdots h(x_k)^{t_k} = 1_G$ since $h(x_1)^{t_1} h(x_2)^{t_2} \cdots h(x_k)^{t_k} = 1_G$. Thus, $a \in \ker(\phi)$.

By choice of N, $N \leq \ker(\phi)$, we'll attempt to define $\tau : \overline{F} \to G$ by $\tau(wN) = \phi(w)$ for $w \in F$.
τ is well defined because $w_1 N = w_2 N \implies w_1^{-1} w_2 \in N \implies w_1^{-1} w_2 \in \ker(\phi) \implies \phi(w_1)^{-1} \phi(w_2) = 1 \implies \phi(w_1) = \phi(w_2)$. Showing τ is a homomorphism is straightforward.

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[dr, "h"'] & F \arrow[d, "\phi_N"] \arrow[dr, "\phi"] & \\ & \overline{F} \arrow[r, dashed, "\mathcal{T}"] & G \end{tikzcd}$$

$\square$

10 Lecture 10: Presentation of Groups (Feb 4)

Example 10.1. *Dihedral Group: D_n = symmetry group of a regular n-gon.*
Let $D_n = \langle a, b\rangle$ where a = rotation by $\frac{2\pi}{n}$ and b = reflection about the vertical axis. Then D_n is generated by a and b with the relations $a^n = 1$ and $b^2 = 1$.
We have $bab^{-1}a = 1$
Let $X = \{x, y\}$ and F be free group and $A = \{x^n, y^2, yxy^{-1}x\}$ and N = least normal subgroup $\supseteq A$. Let $\overline{x} = xN$ and $\overline{y} = yN$. Then $F/N = \langle \overline{x}, \overline{y}\rangle$ and

$$\overline{yxy^{-1}x} = \overline{yxy}^{-1}\overline{x}^{-1} = 1_{\overline{F}}.$$

There is a unique homomorphism $\tau : F/N \to D_n$ such that $\tau(\overline{x}) = a$ and $\tau(\overline{y}) = b$ as $D_n = \langle a, b\rangle$ so τ is surjective. So $|F/N| \geq 2n$.
Claim: $|F/N| \leq 2n$ (so τ is isomorphism)
Proof: Using relations from before, we can simply write any expression in $\overline{x}, \overline{y}$ down to $\overline{x}^s\overline{y}^t$ for $0 \leq s \leq n$ and $0 \leq t \leq 2$.

11 Lecture 11: Category Theory (Feb 6)

11.1 Categories

Example 11.1. *1. Category of sets, if we have two sets A and B then $A \to B$ is a function $f : A \to B$*
2. Category of groups, if we have two groups G and H then $G \to H$ is a homomorphism $\phi : G \to H$
3. Category of topological spaces, if we have two topological spaces X and Y then $X \to Y$ is a continuous function $f : X \to Y$

Definition 11.2. *A category $\mathcal{C}$ consists of the following data:*
1. *A collection of objects $Ob(\mathcal{C})$*
2. *A collection of morphisms $Mor(\mathcal{C})$*
3. *For each morphism f, there are objects $\mathrm{dom}(f)$ and $\mathrm{cod}(f)$ such that $dom(f) \to cod(f)$ is a morphism in $Mor(\mathcal{C})$*
4. *A composition operation $Mor(\mathcal{C}) \times Mor(\mathcal{C}) \to Mor(\mathcal{C})$*
5. *An identity morphism $1_A : A \to A$ for each object $A \in Ob(\mathcal{C})$*

Lemma 11.3. *The following are true:*
1. *If $a \xrightarrow{f} b$ then $f1_a = f$ and $1_b f = f$.*
2. *If $a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d$ then $h(gf) = (hg)f$*

Example 11.4. *Posets: $\mathbb{P}$ is a poset with $\leq$ binary relation such that*
1. *$a \leq a$ for all $a \in \mathbb{P}$*
2. *$a \leq b$ and $b \leq a \implies a = b$*
3. *$a \leq b$ and $b \leq c \implies a \leq c$*

Given poset $\mathbb{P}$, make "poset category" with objects the elements of $\mathbb{P}$. There is only one morphism $a \to b$ if $a \leq b$ and no morphism if $a \not\leq b$.

Definition 11.5. *Let $\mathcal{C}$ be a category. An object is an initial if for all object b there is a exactly one arrow $a \to b$.*

Definition 11.6. *An arrow $a \xrightarrow{f} b$ in $\mathcal{C}$ is an isomorphism iff there is another $b \xrightarrow{f} a$ such that $gf = 1_b$ and $fg = 1_a$.*

Lemma 11.7. *If a, b are initial objects in $\mathcal{C}$ they are uniquely isomorphic.*

Proof. Let $a \xrightarrow{f} b$ and $b \xrightarrow{g} a$ be unique arrows from a to b and b to a respectively. Then we have $a \xrightarrow{gf} a$ and $b \xrightarrow{fg}$ so $gf = 1_a$ and $fg = 1_b$. □

Definition 11.8. *Object b is terminal iff for all a there is a unique $a \to b$ iff b is initial in $\mathcal{C}^{op}$*

Lemma 11.9. *Two terminal objects are uniquely isomorphic.*

Definition 11.10. *Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A functor from $\mathcal{C}$ to $\mathcal{D}$ is $\mathcal{C} \xrightarrow{F} \mathcal{D}$. Assign to each object a of $\mathcal{C}$ an object $F(a)$ of $\mathcal{D}$. For each arrow $a \xrightarrow{f} b$ there is an arrow $F(a) \xrightarrow{F(f)} F(b)$ with $f(1_a) = 1_{F(a)}$. Additionally, $F(gf) = F(g)F(f)$*

12 Lecture 12: Ring Theory (Feb 8)

Definition 12.1. *A ring is a set R with two binary operations $+$ and $\times$ such that*
1. *$(R, +)$ is an abelian group*
2. *$(R, \times)$ is associative*
3. *$a \times (b + c) = a \times b + a \times c$ and $(b + c) \times a = b \times a + c \times a$ for all $a, b, c \in R$*

Example 12.2. *The following are rings*
1. $\mathbb{Z}, 2\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
2. $M_2(\mathbb{Z})$
3. $\{f : \mathbb{R} \to \mathbb{R}$ *continuous*$\}$
4. $\{f : \mathbb{R} \to \mathbb{R}$ *differentiable*$\}$

Definition 12.3. *R is commutative iff $rs = sr$ for all $rs \in R$*

Definition 12.4. *R is unital iff there exists $1_R \in R$ such that $1_R r = r 1_R = r$ for all $r \in R$*

Definition 12.5. *Let R, S be rings. A ring homomorphism from R to S if $\phi : R \to S$ satisfy:*
1. *$\phi(r + s) = \phi(r) + \phi(s)$ for all $r, s \in R$*
2. *$\phi(rs) = \phi(r)\phi(s)$ for all $r, s \in R$*
3. *$\phi(1_R) = 1_S$*

Definition 12.6. *Let S be a subring of R if S is a ring and the inclusion map $i : S \to R$ is a ring homomorphism.*

Definition 12.7. *Let $\phi : R \to S$ be a ring homomorphism then*
1. *$\ker(\phi) = \{r \in R : \phi(r) = 0\}$ is a subring of R*
2. *$\phi(R) = \{s \in S : \exists r \in R, \phi(r) = s\}$ is a subring of S*

Remark 12.8. *In general, not the case that $\ker(\phi)$ is a subring of R.*

Definition 12.9. *Let R be a ring. An ideal of R is a subring I of R such that*
1. *$I \leq (R, +)$*
2. *For all $a \in R$ and $b \in I$, $ab \in I$.*

Definition 12.10. *Let I be an ideal of R then we define $r + I = \{r + i : i \in I\}$*

13 Lecture 13: Ideals and Quotient Rings (Feb 10)

Theorem 13.1. *First Isomorphism Theorem of Rings: Let R, S be rings and $\phi : R \to S$ be a homomorphism with $\ker(\phi)$ an ideal of R and $\mathrm{im}(\phi)$ a subring of S. Then we get isomorphism*

$$\tau : \frac{R}{\ker(\phi)} \xrightarrow{\cong} \mathrm{im}(\phi) \text{ and } \tau : r + \ker(\phi) \mapsto \phi(r).$$

Theorem 13.2. *The following are true for any ideals I, J of R:*

1. *$I + J =$ smallest ideal containing I and J*
2. *$I \cap J =$ largest ideal contained in I and J*
3. *$IJ = \{\sum_{i=1}^{n} r_i s_i | n \in \mathbb{N}, r_i \in I, s_i \in J\} \subseteq I \cap J$*
4. *Ideals of R/I are in bijection with ideals of R that contain I*

Definition 13.3. *R is a zero ring if $R = \langle 0_R \rangle$*
Note: If $1_R = 0_R$ then $r \cdot 1 = r \cdot 0 = 0$ and R is a zero ring.

Definition 13.4. *R is an integral domain iff*

1. *$1_R \neq 0_R$ (R is not a zero ring)*
2. *For all $a, b \in R$*

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

Definition 13.5. *R is a field iff*

1. *$1_R \neq 0_R$*
2. *Every element of $R \setminus \{0_R\}$ has a multiplicative inverse*

Exercise 13.6. *If R is a field then R is an integral domain.*

Definition 13.7. *Let R be a ring. Then $u \in R$ is a unit iff u has a multiplicative inverse.*
We let

$$U(R) = \{u \in R : u \text{ is a unit}\}.$$

$U(R)$ is a group under multiplication.

Definition 13.8. *An ideal I of R is principal iff $I = aR$*

Lemma 13.9. *Let R be any ring then R is the largest ideal and $\{0_R\}$ is the smallest ideal.*

Remark 13.10. *We have $R/R = \{0_R\}$ and $R/\{0_R\} \simeq R$*

Theorem 13.11. *Let R be a field and let I be an ideal then $I = 0$ or $I = R$*

Proof. Assume $I \neq 0$, let $a \in I$ and $a \neq 0$. There exist $b \in R$ such that $ab = 1 \in I$. In

any ring $1 \in I$ iff $I = R$.

$\square$

Theorem 13.12. *If $R \neq 0$ are the only two ideals of R then R is a field.*

Proof. Let $a \in R, a \neq 0$ then $(a) \neq \emptyset$ as $a \in (a)$ so $(a) = R$ so $1 \in (a)$ and $1 = ab$ for some b. $\square$

Definition 13.13. *An ideal I of R is maximal iff*

1. *$I \neq R$*
2. *For all ideals $J \supseteq I$, $J = I$ or $J = R$*

I is maximal in poset $\{J : J \text{ ideal}, J \neq R\}$ ordered by $\subseteq$

Lemma 13.14. *If I is an ideal of R,*

$$I \text{ maximal} \iff R/I \text{ is a field.}$$

Proof. $R \neq I \iff 1_{R/I} \neq 0_{R/I}$. By I maximal iff there are only two ideals of R/I which are 0 and R/I. $\square$

Definition 13.15. *Ideal I is prime iff*

1. *$I \neq R$*
2. *For all $a, b \in R$, $ab \in I \implies a \in I$ or $b \in I$*

Lemma 13.16. *R/I is an integral domain iff I is a prime ideal.*

Definition 13.17. *$Spec(R) = \{I : I \text{ is a prime ideal of } R\}$*

Theorem 13.18. *Let R be a ring. I is an ideal of R, $I \neq R$. Then there is a maximal ideal J such that $I \subseteq J$.*

14 Lecture 14: Zorn's Lemma and Modules (Feb 13)

Lemma 14.1. *Zorn's Lemma: If $\mathbb{P}$ is a poset such that every chain in $\mathbb{P}$ has an upper bound, then for every element p of $\mathbb{P}$ there is a $q \geq p$ with q maximal.*

Example 14.2. *Claim: $\mathbb{Q}$ satisfies the hypothesis of Zorn's Lemma.*

Proof. Let $(I_a : a \in A)$ be a chain in $\mathbb{Q}$. That is, I_a is ideal with $I_a \neq R$ and for $a, b \in A, I_a \subseteq I_b$ or $I_b \subseteq I_a$. We need upper bound, let

$$J = \bigcup_{a \in A} I_a = \{r : \text{there is } a \in A, r \in I_a\}.$$

Verify J is an ideal and for $r, s \in J$ then $r \in I_a$ and $s \in I_b$. WLOG let $I_a \subseteq I_b$ so $r, s \in I_b$ so $r + s \in I_b \subseteq J$.
Verify $J \neq R$, easy as $1 \notin I_a$ all a so $1 \notin J$. □

14.1 Modules

Definition 14.3. *Let R be a ring. A R-module is an abelian group $(M, +)$ equipped with a function $R \times M \to M$ by $(r, m) \mapsto rm$ such that*

1. $0_R m = 0_M, 1m = m$ *for all* $m \in M$
2. $(r_1 + r_2) = r_1 m + r_2 m$ *and* $m(r_1 + r_2) = mr_1 + mr_2$
3. $r(sm) = (rs)m$ *for* $r, s \in R$ *and* $m \in M$

Definition 14.4. *If M is a R-module, we can form "linear combinations"*

$$\sum_{i=1}^{n} r_i m_i \text{ for } r_i \in R \text{ and } m_i \in M.$$

Example 14.5. *Let $R = \mathbb{Z}$ and $(G, +)$ be an abelian group. Then G is a $\mathbb{Z}$-module.*

Definition 14.6. *If M is a R-module, a submodule of M is N such that $N \leq (M, +)$, $rn \in N$ for $r \in R$ and $n \in N$.*

Remark 14.7. *If R is a ring, we can view R as a R-module. The ideals of R are the submodules of R (viewed as a R-module).*

Definition 14.8. *Let M, N be R-modules. A R-module homomorphism is a function $\phi : M \to N$ such that*

1. *ϕ is a group homomorphism for $(M, +)$ to $(N, +)$*
2. $\phi(rm) = r\phi(m)$ *for* $r \in R$ *and* $m \in M$

Note: This is like linear transformation in linear algebra.

15 Lecture 15: Modules (Feb 16)

Definition 15.1. *Let M and N be R-modules then we say $\phi : M \to N$ is R-linear if*

$$\phi\left(\sum_{i=1}^{n} r_i m_i\right) = \sum_{i=1}^{n} r_i \phi(m_i) \text{ for } r_i \in R \text{ and } m_i \in M.$$

Definition 15.2. *Let $M \leq N$ (M is submodule of N), $N/M = \{n + M : n \in N\}$. We define*

1. $(n_1 + M) + (n_2 + M) = (n_1 + n_2) + M$
2. $r(n + M) = (rn) + M$

We can define the R-linear quotient map $\phi_M : N \to N/M$ by $\phi : n \mapsto n + M$ with $\ker(\phi_M) = M$.

Theorem 15.3. *First Isomorphism Theorem for Modules: Let $\phi : M_1 \to M_2$ linear, then*

$$\mathrm{im}(\phi) \simeq \frac{M_1}{\ker(\phi)}.$$

via $m + \ker(\phi) \mapsto \phi(m)$

Definition 15.4. *Given R-module M and set $X \subseteq M$, the least submodule containing X is*

$$\mathrm{Span}(X) = \left\{\sum_{i=1}^{n} r_i x_i | n \in \mathbb{N}, r_i \in R, x_i \in X\right\}.$$

We specifically define $\mathrm{Span}(\emptyset) = \{0_M\}$

Definition 15.5. *If R is a ring, $X \subseteq R$ we write $\langle X \rangle$ for the least ideal containing X*

$$\langle X \rangle = \left\{\sum_{i=1}^{n} r_i x_i | n \in \mathbb{N}, r_i \in R, x_i \in X\right\}.$$

Definition 15.6. *A ring R is a principal ideal ring iff all ideals of R are principal (a principal ideal is of form Ra for some $a \in R$).*

Definition 15.7. *R is a principal ideal domain iff R is integral domain and principal ideal ring.*

Definition 15.8. *A ring R is noetherian iff every ideal of R is finitely generated. (An ideal is finitely generated if $I = Ra_1 + \cdots + Ra_n$ for some $a_1, \ldots, a_n \in R$.)*

Theorem 15.9. *If I is an ideal of R the ideals R/I are in bijection with $\{J : J \text{ ideal of } R, J \supseteq I\}$.*
Also prime ideals of R/I correspond to prime ideals of R that contain I.

Example 15.10. *Common PID's*
1. $\mathbb{Z}$
2. $k[x]$ *for* k *a field*

Definition 15.11. *R is a Euclidean Domain iff*
1. *R is integral domain*
2. *There is a function $\phi : R \setminus \{0\} \to \mathbb{N}$ such that for all $a, b \in R$, for $b \neq 0$, there exist q, r such that $a = bq + r$ either $r = 0$ or $\phi(r) < \phi(b)$*

Example 15.12. *If $R = \mathbb{Z}$ then $\phi(n) = |n|$ is a Euclidean Domain.*

Example 15.13. *If $R = k[x]$ for k a field then $\phi(f) = \deg(f)$ is a Euclidean function.*

Theorem 15.14. *If R is a Euclidean Domain then R is a PID.*

Proof. Let R be euclidean domain and let I be the ideal of R. If $I = 0$ then we're done. If $I \neq 0$ consider $\{\phi(r) | r \in I, r \neq 0\} \subseteq \mathbb{N}$. Let n be the minimal element and let $a \in I \setminus \{0\}$ such that $\phi(a) = n$. As $a \in I, Ra \subseteq I$. Let $b \in I$, find q, r such that $b = aq + r$ either $r = 0$ or $\phi(r) < n$. We have $r = b - aq \in I$. So $r = 0$ and $b \in Ra$. $\square$

Lemma 15.15. *Facts about Integral Domains: Let R be an integral domain.*
1. *If $a \neq 0$ and $ab = ac$ then $a(b - c) = 0$*
2. *$Ra = Rb$ iff a and b are associates*

Lemma 15.16. *Let R be any ring, the following are true about units*
1. *u unit iff u has a multiplicative inverse*
2. *Units of R form a group under multiplication*
3. *$a, b \in R$ are called associates iff there is an unit u such that $au = b$.*

Example 15.17. *$\mathbb{Z}[x]$ is not a PID. Consider the ideal $\langle 2, x \rangle = \{f :$ constant term of f is even$\}$.*

Theorem 15.18. *R Noetherian iff $R[x]$ is Noetherian.*

16 Lecture 16: Integral Domains

Definition 16.1. *Let $p \in R$*

1. *p is prime if and only if p is a nonzero non-unit and for all $r, s \in R$, $p|rs$ implies $p|r$ or $p|s$.*
2. *p is irreducible if and only if p is a nonzero non-unit and for all $r, s \in R$ and $p = rs$ then one of r or s is a unit.*

Lemma 16.2. *The following are true*

1. *As R is an ID, for any $a \in R$, a nonzero non-unit iff $(a) \notin \{0, R\}$*
2. *If p is prime, then (p) is a prime ideal*
3. *p prime implies p is irreducible.*
4. *If R PID then prime = irreducible.*
5. *In any ID, (0) is a prime ideal.*

Definition 16.3. *R is a unique factorization domain (UFD) iff*

1. *R is an integral domain*
2. *Every non-zero nonunit in R is a product $u_1, ..., u_n$, $n > 0, u_i$ irreducible.*
3. *If $m, n > 0$ then $u_1...u_n = v_1...v_m$ and u_i, v_j irreducible, then $m = n$ and u_i are associates and permutations of v_j*

Recall 16.4. *An R-module M is finitely generated (fg) if there are $m_1, ..., m_n \in M$ such that $M = \mathrm{Span}(m_1, ..., m_n)$.*

Definition 16.5. *An R-module N is Noetherian iff every submodule of N is finitely generated.*
Equivalently, Ring R is Noetherian iff R is Noetherian as an R-module that is all ideals are finitely generated.

Example 16.6. *Non-Noetherian Rings:*

$$R = \bigcup_{n=1}^{\infty} \mathbb{Z}[x_1, ..., x_n].$$

Consider the ideal

$$I = \langle x_1, x_2, ... \rangle = \{f \in R : f \text{ has no constant term}\} = \{f : f(0, ...) = 0\}.$$

Proof. If I is finitely generated there is $h_1, ..., h_n \in I$ such that

$$I = Rh_1 + \cdots + Rh_n = (h_1) + \cdots + (h_n).$$

Each h_i has finite x_j's in it so find N so large that x_N does not appear in any h_i.
As $x_N \in I$ there are $f_1, ..., f_n \in R$ such that $x_N = \sum_{i=1}^{n} f_i h_i$. Substitute

$$x_j = \begin{cases} 0 & \text{if } j < N \\ 1 & \text{if } j \geq N \end{cases}.$$

Then we conclude $1 = 0$ which is a contradiction. $\square$

Theorem 16.7. *Let M be a R-module then the following are equivalent*

1. *M is Noetherian*
2. *For any increasing change of submodules $N_0 \subseteq N_1 \subseteq N_2 \subseteq \cdots$ of M, the chain is eventually constant.*
3. *For any nonempty set X of submodules of M, there is some $N \in X$ which is maximal under inclusion.*

Proof. (1) $\implies$ (2): Consider $N_\infty = \bigcup_{i=0}^{\infty} N_i = \{m : \exists i \in \mathbb{N}, m \in N_i\}$.
Verify that N_∞ is a submodule of M, as M is noetherian, N_∞ is finitely generated say $N_\infty = \operatorname{Span}(X)$ with $X \subseteq N_\infty$ finite. Find i such that $X \subseteq N_i$ then $N_\infty = \operatorname{Span}(X) \subseteq N_i \subseteq N_j \subseteq N_\infty$ for $j \geq i$.

(2) $\implies$ (1): AFSOC there is $N \leq M$ with N not finitely generated. Take $n_0 \in N$ and $n_0 \neq 0$. Choose $n_1 \in N \setminus Rn_0$ and generally

$$n_k \subseteq N \setminus \left(\sum_{i<k} Rn_i \right).$$

In the end, let $N_i = \sum_{j<i} Rn_j$ then $N_0 < N_1 < N_2 < \cdots$, a contradiction. $\square$

17 Lecture 17: PID (Feb 19)

Lemma 17.1. *If R PID then*
1. *R is Noetherian*
2. *0 is a prime ideal and 0 is maximal iff R is a field*
3. *For $a \neq 0, (a)$ prime iff a prime iff (a) maximal iff a irreducible*

Theorem 17.2. *If R is a PID then R UFD*

Claim 17.3. *Every nonzero non-unit $a \in R$ is a finite product of irreducibles.*

Proof. If not, choose a that is not a finite product of irreducibles such that (a) is maximal under inclusion, a not finite product of irreducibles so a is not irreducible.
So $a = bc$ where b, c not units. So $(a) \subsetneq (b)$ and $(a) \subsetneq (c)$ so b, c are finite products of irreducibles as (a) is maximal under inclusion. Contradiction so a is a finite product of irreducibles. □

Claim 17.4. *If $u_1...u_n = v_1...v_n$ with u_i, v_j irreducible then $m = n$ and u_i, v_j are the same up to permutations and associates.*

Proof. u_1 divides $v_1...v_m$. As R PID, u_1 is prime so $u_1|v_j$ for some j. As v_j is irreducible, u_i either unit or an associate of v_j. As u_i irreducible, u_i not a unit. So u_1 and v_j are associates. Say $u_1 = x \cdot v_j$ for x unit. Then

$$xu_2...u_n = v_1...v_{j-1}v_{j+1}...v_m.$$

Finish by induction. □

Corollary 17.5. *If k field, $k[x]$ is a UFD.*

Lemma 17.6. *If R PID and $p \in R$ irreducible then $R/(p)$ is a field.*

Theorem 17.7. *(Hilbert's Basissatz Theorem) If R Noetherian ring then $R[x]$ is Noetherian ring.*

Proof. Let I be ideal of $R[x]$. For each n let

$$J_n = \left\{ r \in R : \exists c_0, ..., c_{n-1} \in R, rx^n + \sum_{i=0}^{n-1} c_i x^i \in I \right\}.$$

J_n is an ideal of R.
Claim: $J_n \subseteq J_{n+1}$
Let $r \in J_n$ say $rx^n + \sum_{i=0}^{n-1} c_i x^i \in I$, as I ideal, $x(rx^n + ...) \in I$ so $r \in J_{n+1}$.

As R is noetherian, there is N such that $J_n = J_N$ for all $n \geq N$.
For each $n \leq N$, as R noetherian, J_n is finitely generated ideal of R. For each $n \leq N$,

choose a finite set $F_n \subseteq I$ such that if $\deg(f) = n$ then $f \in F_n$ and $\{$leading coeff of f : $f \in F_n\}$ generates J_n.
Let $F = \bigcup_{n \leq N} F_n$ and claim that F generates I (as an ideal of $R[x]$). Show by induction on $\deg(h)$ that every $h \in I$ is an $R[x]$-linear combination of elements of F.
(Base Case) $h = 0$, $\deg(h) \leq N$ and $h = rx^n + \sum_{in} c_i x^i$ where $r \in J_n$. SO find $\lambda_1, ..., \lambda_t \in R$ and $g_1, ..., g_t \in F_n$ such that $r =$ leading coefficient of $\sum_{i=1}^{t} \lambda_i g_i$. Look at $h - \sum_{i=1}^{t} \lambda_i g_i \in I$ has lower degree.
(Induction Step) Let $h \in I$ with $\deg(h) = n > N$, $h = rx^n + \sum_{i<n} c_i x^i$ with $r \in J_n = J_N$. Find $g'_1, ..., g'_s \in F_N, \lambda_1, ..., \lambda_s \in R$ such that $r =$ leading coefficients of $\sum_{j=1}^{s} \lambda_j g'_j$ with leading term in rx^N. So now look at $h - \sum_{j=1}^{s} \lambda_j x^{n-N} g'_j$. This has degree $< n$, so done by induction. $\square$

18 Lecture 18: Unique Factorization Domains (Feb 23)

Theorem 18.1. *R UFD $\implies$ $R[x]$ UFD*

Definition 18.2. *Field of Fractions: Given any integral domain R, constructing its "field of fractions" as $X = \{(a,b) \in R \times R : b \neq 0\}$.*
Define binary relation $\sim$ on X by

$$(a_1, b_1) \sim (a_2, b_2) \iff a_1 b_2 = a_2 b_1.$$

Definition 18.3. *Define $\frac{a}{b}$ to be the $\sim$-equivalence class of (a,b).*
The fractions form a ring F with $1_F = \frac{1}{1}$ and $0_F = \frac{0}{1}$. Every element has an inverse, so F is a field.

Remark 18.4. *If R is integral domain and F is a field of fractions then $F[x]$ is a PID.*

Definition 18.5. *Let R be an integral domain and let $a_1, ..., a_n \in R, a_i \neq 0$. A greatest common divisor (gcd) for $a_1, ..., a_n$ is $a \in R$ such that*
1. *$a | a_i$ for $i = 1, ..., n$*
2. *If $b | a_i$ for $i = 1, ..., n$ then $b | a$*

Remark 18.6. *If $\gcd(a_1, ..., a_n)$ exists, it is unique up to associates.*

Lemma 18.7. *If R UFD, then gcd's exist.*

Lemma 18.8. *If R PID and $(a_1, ..., a_n) = Ra_1 + \cdots + Ra_n$ then*

$$(a_1, ..., a_n) = (d) \iff d = \gcd(a_1, ..., a_n).$$

Definition 18.9. *Let R, S be rings, $\phi : R \to S$ a ring homomorphism.*
1. *If I is an ideal of R, I^e (extension of I) = ideal generated by $\phi[I]$ in S.*
2. *If J ideal of S, J^c (contraction) of J is $\phi^{-1}[J] = \{r \in R : \phi(r) \in J\}$.*

Lemma 18.10. *If we define $\psi : R/J^c \to S/J$ by $\psi : r + J^c \mapsto \phi(r) + J$, ψ is well defined injective homomorphism.*

Lemma 18.11. *If J is a prime ideal of S, then J^c is a prime ideal of R.*

Proof. S/J is ID, so all subrings of S/J are ID's, so R/J^c is ID, so J^c is prime ideal. $\square$

Recall 18.12. *If R a ring, a subset $S \subseteq R$ is multiplicatively closed iff $1 \in S$ and for all $s, t \in S$, $st \in S$.*
Define a "ring of fractions" $S^{-1}R$.

Theorem 18.13. *Let $X = \{(a, b) : a \in R, b \in S\}$ then*

$$(a_1, b_1) \sim (a_2, b_2) \iff \exists s \in S, s(a_1b_2 - a_2b_1) = 0.$$

Theorem 18.14. *Let the set be $S^{-1}R$. Define the map $\phi : R \to S^{-1}R$ by $\phi : r \mapsto \frac{r}{1}$. Then ϕ is a ring homomorphism but it may not be injective.*

Theorem 18.15. *Universal Property: For all rings A and all homomorphism $\psi : R \to A$ such that $\phi(s)$ is a unit in A for all $s \in S$. There is unique $X : S^{-1}R \to A$ such that $\psi = X \circ \phi$.*

$$\begin{tikzcd} R \arrow[r, "\phi"] \arrow[dr, "\psi"'] & S^{-1}R \arrow[d, dashed, "X"] \\ & A \end{tikzcd}$$

Proof. $X\left(\frac{x}{s}\right) = \psi(r)\psi(s)^{-1}$ □

19 Lecture 19: (Feb 25)

Theorem 19.1. *R UFD $\implies$ $R[x]$ UFD*

Remark 19.2. *As R ID for nonzero, $f, g \in R[x]$, $\deg(fg) = \deg(f) + \deg(g)$*

Definition 19.3. *Let $f \in R[x]$ with $f = 0$, f is primitive iff there is no prime $p \in R$ such that $p|f$ in $R[x]$*

Remark 19.4. *Informally, the* gcd *of the coefficients of f is 1.*

Lemma 19.5. *(Gauss's Lemma) If $f, g \in R[x]$ both primitive then fg is primitive.*

Proof. Suppose not, and let p be prime in R such that $p|fg$. As p prime, $P = (p)$ is prime ideal so R/P is ID. Let $\phi : R \to R/P$ quotient map.
Define $\phi^* : R[x] \to R/P[x]$ by

$$\phi^*\left(\sum_{i=0}^{n} a_i x^i\right) = \sum_{i=0}^{n} \phi(a_i) x^i.$$

ϕ^* is homomorphism. So $\phi^*(fg) = \phi(f)\phi(g)$ but $\phi^*(fg) = 0$ as $p|fg$.
As R/P ID, $\frac{R}{P}[x]$ is ID so $\phi^*(f) = 0$ or $\phi^*(g) = 0$. So $p|f$ or $p|g$ so a contradiction. □

Lemma 19.6. *Let $f \in R[x]$, $f \neq 0$. Then there are $C \in R$, $g \in R[x]$, g primitive and $f = Cg$*

Proof. $C = \gcd(a_0, ..., a_n)$ if $f(x) = a_0 + a_1 x + \cdots + a_n x^n$. □

Lemma 19.7. *Let $h \in F[x]$, then there is a nonzero $a \in F$ such that $ah \in R[x]$ and ah primitive.*

Lemma 19.8. *Let $f \in R[x]$ be irreducible with $\deg f > 0$. Then f irreducible in $F[x]$*

Proof. Proof of Theorem 19.1:
To show existence of factorization: Let $f \in R[x]$ nonzero and nonunit. Write $f = cg$ where $c \in R$ and $g \in R[x]$ primitive. If C nonunit (in R), use R being UFD to write $c = r_1 ... r_n$ with r_i irreducible in R.
Note: r_i still irreducible in $R[x]$.
We have $g \in R[x] \implies g \in F[x]$. If $\deg(g) > 0$ then g is not a unit and using $F[x]$ is UFD to write

$$g = h_1 ... h_n \text{ where } h_i \in F[x], h_i \text{ irreducible in } F[x].$$

Find $a_i \neq 0$ in F such that $H_i = a_i h_i \in R[x]$ primitive.
Note: H_i irreducible in $F[x]$.

So we can write

$$\prod_{i=1}^{n} H_i = \left(\prod_{i=1}^{n} a_i\right)\left(\prod_{i=1}^{n} h_i\right) = \left(\prod_{i=1}^{n} a_i\right) g.$$

Let $\prod a_i = \frac{A}{B}$ for $A, B \in R$ and $A, B \neq 0$. Then $B(\prod H_i) = Ag$. Argue that A, B must be associates so $\prod a_i$ unit of R and WLOG $\prod a_i = 1$.

To show uniqueness of factorization:
Suppose $f = r_1 ... r_n g_1 ... g_s = r'_1 ... r'_m g'_1 ... g'_s$ for $r_i, r'_i \in R$ irreducible and $g_i, g'_i \in R[x]$ irreducible and $\deg(g_i), \deg(g'_i) > 0$ for all i.
As g_i, g'_i are primitive, $g_1 ... g_s$ and $g'_1 ... g'_s$ are primitive.
So $v_1 ... v_n, v'_1, ..., v'_m$ are associates in R and using R being UFD. By lemma 19.8, g, g' are irreducible in $F[x]$. $\square$

20 Lecture 20: (Mar 2)

Definition 20.1. *Let $\mathcal{C}$ be a category and c, d objects then a cone over c, d is a diagram*

$$\begin{tikzcd} & e \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ c & & d \end{tikzcd}$$

A cone over c, d is a product of c, d iff for all cones

$$\begin{tikzcd} & e' \arrow[dl, "\alpha'"'] \arrow[dr, "\beta'"] & \\ c & & d \end{tikzcd}$$

there is a unique $\phi : e' \to e$ such that the diagram

$$\begin{tikzcd} & e' \arrow[dl, "\alpha'"'] \arrow[d, dashed, "\phi"] \arrow[dr, "\beta'"] & \\ c & e \arrow[l, "\alpha"] \arrow[r, "\beta"'] & d \end{tikzcd}$$

commutes (i.e. $\alpha \circ \phi = \alpha'$ and $\beta \circ \phi = \beta'$).

Example 20.2. *Groups: Let G_1, G_2, H be groups and $\phi : H \to G_1$, $\psi : H \to G_2$ homomorphisms. The product cone is $(G_1 \times G_2, \pi_1, \pi_2)$ where $\pi_1(g_1, g_2) = g_1$ and $\pi_2(g_1, g_2) = g_2$. The unique $\rho : H \to G_1 \times G_2$ with $\rho(h) = (\phi(h), \psi(h))$ makes the diagram*

$$\begin{tikzcd} & H \arrow[dl, "\phi"'] \arrow[d, dashed, "\rho"] \arrow[dr, "\psi"] & \\ G_1 & G_1 \times G_2 \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & G_2 \end{tikzcd}$$

commute ($\pi_1 \circ \rho = \phi$ and $\pi_2 \circ \rho = \psi$).

Recall 20.3. *For any category $\mathcal{D}$, a terminal object is d object such that all objects $\overline{d}$ there is exactly one $\overline{d} \to d$.*

Lemma 20.4. *Given $\mathcal{C}$ and objects c, d make a new category whose objects are the cones over c, d.*

Cone 1

$$\begin{tikzcd} & e \arrow[dl, "\alpha"'] \arrow[dr, "\beta"] & \\ c & & d \end{tikzcd}$$

Cone 2

$$\begin{tikzcd} & e' \arrow[dl, "\alpha'"'] \arrow[dr, "\beta'"] & \\ c & & d \end{tikzcd}$$

Drawing the arrow from cone 1 to cone 2 we have

$$\begin{tikzcd} & e \arrow[dl, "\alpha"'] \arrow[d, dashed, "\phi"] \arrow[dr, "\beta"] & \\ c & e' \arrow[l, "\alpha'"] \arrow[r, "\beta'"'] & d \end{tikzcd}$$

such that $\alpha \circ \phi = \alpha'$ and $\beta \circ \phi = \beta'$.
Being a product of c, d is equivalent to being a terminal object in category of cones over c, d

Definition 20.5. *A co-cone over c, d is a diagram*

$$\begin{tikzcd} c \arrow[dr, "\alpha"'] & & d \arrow[dl, "\beta"] \\ & e & \end{tikzcd}$$

is a coproduct of c, d iff for all cocones

$$\begin{tikzcd} c \arrow[dr, "\alpha'"'] & & d \arrow[dl, "\beta'"] \\ & e' & \end{tikzcd}$$

there is a unique $\phi : e \to e'$ such that the diagram

$$\begin{tikzcd} c \arrow[dr, "\alpha"] \arrow[ddr, "\alpha'"'] & & d \arrow[dl, "\beta"'] \arrow[ddl, "\beta'"] \\ & e \arrow[d, dashed, "\phi"] & \\ & e' & \end{tikzcd}$$

commutes (i.e. $\alpha' \circ \phi = \alpha$ and $\beta' \circ \phi = \beta$).

Example 20.6. *Sets: Let X, Y, Z be sets then define $X \cup Y = (X \times \{0\}) \cup (Y \times \{1\})$. Define $i : x \mapsto (x, 0)$ and $j : y \mapsto (y, 1)$. Then the coproduct is $h(x, 0) = f(x)$ and $h(y, 1) = g(y)$.*

$$\begin{tikzcd} X \arrow[d, "i"] \arrow[dr, "g"] & Y \arrow[d, "f"] \arrow[dl, "j"'] \\ X \cup Y \arrow[r, dashed, "h"] & Z \end{tikzcd}$$

such that $h(i(x)) = f(x)$ and $h(j(y)) = g(y)$.

Definition 20.7. *Let M_1, M_2 be R-modules. Define*

$$M_1 \oplus M_2 = \{(m_1, m_2) : m_i \in M_i\}.$$

Define $i : M_1 \to M_1 \oplus M_2$ by $i(m_1) = (m_1, 0)$ and $j : M_2 \to M_1 \oplus M_2$ by $j(m_2) = (0, m_2)$. For any R-module N and homomorphisms $\psi_1 : M_1 \to N$, $\psi_2 : M_2 \to N$, there is a unique $\rho : M_1 \oplus M_2 \to N$ such that the diagram

$$\begin{tikzcd}
M_1 \arrow[d, "i"] \arrow[dr, "\psi_1"] & M_2 \arrow[d, "j"] \arrow[dl, "\psi_2"'] \\
M_1 \oplus M_2 \arrow[r, dashed, "\rho"] & N
\end{tikzcd}$$

commutes ($\rho \circ i = \psi_1$ and $\rho \circ j = \psi_2$). We have $\rho(m_1, m_2) = \rho((m_1, 0) + (0, m_2)) = \psi_1(m_1) + \psi_2(m_2)$.

For the coproducts, $\pi_1 : M_1 \oplus M_2 \to M_1$ by $\pi_1(m_1, m_2) = m_1$ and $\pi_2 : M_1 \oplus M_2 \to M_2$ by $\pi_2(m_1, m_2) = m_2$. For any R-module N and homomorphisms $\psi_1 : N \to M_1$, $\psi_2 : N \to M_2$, there is a unique $\rho : N \to M_1 \oplus M_2$ such that the diagram

$$\begin{tikzcd}
N \arrow[d, "\psi_1"] \arrow[dr, "\psi_2"'] & M_1 \arrow[d, "\pi_1"] \arrow[dl, "\pi_2"] \\
M_1 \oplus M_2 \arrow[r, dashed, "\rho"] & N
\end{tikzcd}$$

commutes ($\rho \circ \pi_1 = \psi_1$ and $\rho \circ \pi_2 = \psi_2$). We have $\rho(n) = \rho(\psi_1(n), \psi_2(n)) = (\psi_1(n), 0) + (0, \psi_2(n))$.

21 March 11

Lemma 21.1. *Fix R-module, M and N consider $Hom_R(M, N) = \{\phi : M \to N : \phi R - linear\}$. This forms an R-module.*

Theorem 21.2. *Given ring R and set X, to construct a R-module on X. Let $M = \{g : X \to R, \{x \in X : g(x) \neq 0\}$ is finite$\}$. To make M an R-module, $(g + h)(x) = g(x) + h(x)$ and $(rg)(x) = r(g(x))$.*
We'll write elements of M as $\lambda_1 x_1 + \cdots + \lambda_n x_n$ where $\lambda_i \in R$ and $x_i \in X$

Claim 21.3. *For any R-module N, for any function $f : X \to N$, there is a unique R-linear $\psi : M \to N$ such that $\psi(1x) = f(x)$ for all $x \in X$.*

Proof.

$$\psi\left(\sum_{i=1}^{n} \lambda_i x_i\right) = \sum_{i=1}^{m} \lambda_i f(x_i).$$

□

Theorem 21.4. *Any module is isomorphic to a quotient of a free module.*

Proof. Let N be arbitrary, let $X =$ underlying set of N, $f : X \to N$ where $f(n) = n$. $\psi : M \to N$ with $M = Free_R(X)$, ψ is surjective. □

Remark 21.5. *When I constructed $Free_R(X)$, every element of $Free_R(X)$ is a unique R-linear combination of elements of elements from the set $\{1x : x \in X\}$.*

Definition 21.6. *R-module M is free iff M has a linearly independent generating set.*

Lemma 21.7. *If B is such a set, $M \simeq Free_R(B)$*

Definition 21.8. *Let M, N, P be R-modules. A function $\phi : M \times N \to P$ is R-bilinear if*

1. *For all n, $m \mapsto \phi(m, n)$ is R-linear*
2. *For all m, $n \mapsto \phi(m, n)$ is R-linear*

Remark 21.9. *Bilinear functions from $M \times N$ to P are very different from linear functions from $M \oplus N$ to P.*

Example 21.10. *$R = \mathbb{R}$ and $M = N = \mathbb{R}'$:*

1. *$R' \times \mathbb{R}' \to \mathbb{R}' : (x, y) \mapsto xy$ is bilinear, not linear*
2. *$R' \oplus \mathbb{R}' = \mathbb{R}' : (x, y) \mapsto x + y$ is linear, not bilinear*

Definition 21.11. *Given M, N R-modules we will construct an R-module $M \otimes_R N$ (tensor product of M and N) together with a bilinear map $\otimes : M \times N \to M \otimes_R N$ by $(m, n) \mapsto m \otimes n$ such that for all R-module P and all bilinear $\phi : M \times N \to P$ there is unique R-linear $\rho : M \otimes_R N \to P$ such that $\phi = \rho \circ \otimes$*

$$\begin{tikzcd} M \times N \arrow[r, "\otimes"] \arrow[dr, "\phi"'] & M \otimes N \arrow[d, dashed, "\rho"] \\ & P \end{tikzcd}$$

commutes.
"$\otimes$ is initial in the category of bilinear maps with domain $M \times N$"

Theorem 21.12. *$Hom_R(M, Hom_R(N, P)) \simeq$ bilinear maps from $M \times N \to P \simeq Hom_R(M \otimes N, P)$*

22 March 11

22.1 Goal and Preliminaries

Remark 22.1 (Note on Bilinearity)**.** *Let $\phi : M \times N \to P$ be a bilinear map. If $\beta : P \to Q$ is a linear map, then the composition $\beta \circ \phi$ is bilinear.*

Definition 22.2. *(Universal Property of Tensor Products)*
Goal: *Construct a module $M \otimes N$ and a bilinear map $\otimes : M \times N \to M \otimes N$ such that for any module Q and any bilinear map $\psi : M \times N \to Q$, there exists a unique linear map $\beta : M \otimes N \to Q$ that makes the following diagram commute:*

$$\begin{tikzcd} M \times N \arrow[r, "\otimes"] \arrow[dr, "\psi"'] & M \otimes N \arrow[d, dashed, "\exists!\beta"] \\ & Q \end{tikzcd}$$

In other words, $\psi = \beta \circ \otimes$.

Lemma 22.3 (Easy Fact on Quotients)**.** *Let $A \leq B$ be submodules. Then:*

1. *The natural projection $\phi_A : B \to B/A$ is a homomorphism, and it is 0 on A.*
2. *For any module C and homomorphism $\gamma : B \to C$ such that $\gamma(A) = 0$, there is a unique linear map $\delta : B/A \to C$ such that the diagram commutes:*

$$\begin{tikzcd} B \arrow[r, "\phi_A"] \arrow[dr, "\gamma"'] & B/A \arrow[d, dashed, "\exists!\delta"] \\ & C \end{tikzcd}$$

Specifically, δ is defined by mapping $b + A \mapsto \gamma(b)$.

22.2 Construction of the Tensor Product

Definition 22.4 (Constructing $M \otimes N$ and $\otimes$)**.** *Let $X = M \times N$ be the Cartesian product. Construct the free R-module on X, denoted $Fr(X)$, whose elements are finite sums of the form:*

$$\sum \lambda_i(m_i, n_i) \quad \text{where } \lambda_i \in R.$$

Consider the map $\sigma : M \times N \to Fr(X)$ defined by $(m, n) \mapsto 1(m, n)$.
For σ to be bilinear, we would need equations like $(m_1 + m_2, n) - (m_1, n) - (m_2, n) = 0$ and $(\lambda m, n) - \lambda(m, n) = 0$ to hold. However, in $Fr(X)$, these are not *true!*
To fix this, let B be the submodule of $Fr(X)$ generated by all elements of the form that "need to be 0" to satisfy bilinearity. Define the tensor product as the quotient module:

$$M \otimes N = Fr(X)/B$$

and define the map $\otimes : M \times N \to M \otimes N$ by:

$$\otimes : (m, n) \mapsto 1(m, n) + B = m \otimes n.$$

Claim 22.5. *The map $\otimes$ is bilinear.*

Proof. We check one of the bilinearity conditions (the others follow similarly).

$$\begin{aligned}\otimes(\lambda m, n) - \lambda \otimes (m, n) &= 1(\lambda m, n) + B - \lambda(m, n) + B \\ &= [1(\lambda m, n) - \lambda(m, n)] + B \\ &= 0 + B\end{aligned}$$

This evaluates to $0 + B$ because the term in the brackets is one of the generators of the submodule B, meaning it lives inside B. The same logic applies for the other three bilinearity conditions. □

Claim 22.6 (Fulfilling the Goal)**.** *The constructed $M \otimes N$ satisfies the universal property. Let $\psi : M \times N \to Q$ be bilinear. There exists a unique linear $\beta : M \otimes N \to Q$.*

Proof. Let $\phi : Fr(X) \to Q$ be the R-module homomorphism defined on the basis by the unique R-linear extension:

$$\phi\left(\sum \lambda_i(m_i, n_i)\right) = \sum \lambda_i \psi(m_i, n_i) \implies \phi(1(m, n)) = \psi(m, n).$$

We claim $\phi(b) = 0$ for all $b \in B$. Since B is generated by the bilinearity relations, we evaluate ϕ on those generators. For instance:

$$\phi((\lambda m, n) - \lambda(m, n)) = \psi(\lambda m, n) - \lambda\psi(m, n) = 0$$

because ψ is explicitly given as bilinear. Since ϕ maps all 4 types of generators of B to 0, $\phi(B) = 0$.
By our "Easy Fact", there exists a unique linear map $\beta : \frac{Fr(X)}{B} \to Q$ such that the diagram commutes:

$$\begin{tikzcd}
M \times N \arrow[r, "\sigma"] \arrow[dr, "\otimes"'] \arrow[ddr, bend right, "\psi"'] & Fr(X) \arrow[d, "\text{quotient}"] \arrow[dd, bend left] \\
& \frac{Fr(X)}{B} = M \otimes N \arrow[d, dashed, "\beta"] \\
& Q
\end{tikzcd}$$

For elements in $M \otimes N$, we have $m \otimes n = (m, n) + B \implies \beta(m \otimes n) = \psi(m, n)$. □

Remark 22.7 (Uniqueness of $M \otimes N$)**.** *Suppose $\otimes' : M \times N \to M \otimes' N$ is another bilinear map satisfying the universal property. Then, any bilinear map is of the form $\delta \circ \otimes'$ for a linear map δ on $M \otimes' N$. By universal properties, there is a unique isomorphism $I : M \otimes N \to M \otimes' N$ such that $I(m \otimes n) = m \otimes' n$ for all m, n.*

22.3 Tensor Product with the Base Ring

Theorem 22.8. *Let M be an R-module. Then $R \otimes_R M \cong M$.*

Proof. Consider the scalar multiplication map $t : R \times M \to M$ defined by $(r, m) \mapsto rm$. For any bilinear $\psi : R \times M \to Q$, we want to show there exists a unique linear map $\beta : M \to Q$ such that $\psi = \beta \circ t$.

$$\begin{tikzcd} R \times M \arrow[r, "t"] \arrow[dr, "\psi"'] & M \arrow[d, dashed, "\exists!\beta"] \\ & Q \end{tikzcd}$$

If β exists, we must have:

$$\beta(m) = \beta(1 \cdot m) = \beta(t(1, m)) = \psi(1, m).$$

This defines β uniquely. We verify two claims:

1. **Claim 1:** β is linear because ψ is bilinear.
2. **Claim 2:** $\beta \circ t = \psi$.
   Proof: $\psi(r, m) = r\psi(1, m) = \psi(1, rm) = \beta(rm) = \beta(t(r, m))$.

By the uniqueness property of the tensor product, we conclude:

1. $R \otimes M \cong M$.
2. There exists a unique isomorphism $I : R \otimes M \to M$ mapping $r \otimes m \mapsto rm$.

$\square$

23 March 16

Definition 23.1. *An R-module M is cyclic iff there exist m such that $M = Rm$*

Definition 23.2. *M is an R-module, $m \in M$, the annihilator of m is $Ann(m) = \{r : rm = 0\}$*

Lemma 23.3. *For any M R-module any $m \in M$, have R-module homomorhpism $\phi : R \to M$, $\phi : r \mapsto rm$.. Then we have $Im(\phi) = Rm$ and $\ker(\phi) = Ann(m)$ so*

$$Rm \simeq \frac{R}{Ann(m)}.$$

Remark 23.4. *If M cyclic, $M = Rm$, $M \simeq \frac{R}{Ann(m)}$*

Remark 23.5. *If we have I ideal of R and view R-module, then*

$$R/I = R(1 + I) \text{ and } Ann(1 + I) = I.$$

Remark 23.6. *Let R be PID, cyclic R-module $\simeq R/(c)$ for $c \in R$.*

Definition 23.7. *Let M be R-module, P, Q submodules of M are complementary if $M = P + Q$ and $P \cap Q = 0$*

Definition 23.8. *If P, a submodule of M, P is a direct summand of M if there is a complementary Q.*

Example 23.9. *The following are examples of direct summand*

1. $R = \mathbb{R}, M = \mathbb{R}^2$
2. $R = \mathbb{Z}, M = \mathbb{Z}/4\mathbb{Z}$, $P = \{0, 2\}$ *then P has no complement*
3. M_i *R-module,* $i = 1, ..., n$, $M_1 \oplus \cdots \oplus M_n = \{(m_1, ..., m_n) : m_i \in M_i\}$ *with obvious operations*
4. *If $P_1, ..., P_n$ submodules of N and $N = P_1 + P_2 + \cdots + P_n$ and $\sum_{i=1}^n p_i = 0$ ($p_i \in P_i$) then all $p_i = 0, N \simeq P_1 \oplus \cdots \oplus P_n$*

Lemma 23.10. *A free module on a finite set $X = \{x_1, ..., x_n\}$ with $|X| = n$ then $Fr_R(X) \simeq R^n$ by*

$$\sum_{i=1}^n \lambda_i x_i \mapsto (\lambda_1, ..., \lambda_n).$$

Definition 23.11. *Let R be ID, M R-module, $m \in M$ is a torsion element if there is $r \neq 0$ such that $rm = 0$.*

Definition 23.12. *R ID, M is R-module, M is torsion free if 0 is the only torsion element.*

Definition 23.13. *If M is torsion free R-module, R ID, then $m_1, ..., m_n$ are linearly independent if they are distinct and*

$$\sum_{i=1}^{n} r_i m_i = 0.$$

Definition 23.14. *R ID, M torsion free R-module. M has rank n iff $n = \max\{|I| : I \subseteq M, I$ independent$\}$.*

Theorem 23.15. *Let R be PID and let M be a finitely generated R-module. Then M is isomorphic to*

$$\frac{R}{(c_1)} \oplus \cdots \oplus \frac{R}{(c_n)}.$$

for c_i's are nonzero nonunit elements of R and c_i divides c_{i+1} Moreover:

1. *t unique*
2. *n unique*
3. *ideals (c_i) are unique equivalently, c_i's are unique up to associates.*

Lemma 23.16. *If R ID, R^n has rank n.*

Proof. Let $e_i = (0, ..., 1, ..., 0)$ where 1 is in the i-th position. We have $e_1, ..., e_n$ are independent. Let $I \subseteq R^n$ be independent. Let k be the field of fraction of ID R. $I \subseteq R^n \subseteq k^n$.
Claim: I is an independent subset of the k-vs k^n. Prove this by clearing fraction.
Then by linear algebra $|I| \leq \dim_k(k^n) = n$.
The result will follow from the following theorem

Theorem 23.17. *If R PID, N free of rank n and $M \leq N$ then M is free of some rank $m \leq n$.*

There is a basis $z_1, ..., z_n$ of N and there are nonzero nonunit $c_1, ..., c_m$ such that c_i divides c_{i+1} and $c_1 z_1, ..., c_m z_m$ forms a basis of M. □

24 March 18

Theorem 24.1. *Let R be a principal ideal domain (PID) and M be a finitely generated R-module. Then*

$$M \simeq \frac{R}{(c_1)} \oplus \cdots \oplus \frac{R}{(c_n)} \oplus R^t$$

where $t \geq 0$, $c_i \neq 0$, $c_i \mid c_{i+1}$ for all $1 \leq i < n$, and the elements c_i are non-units.

Lemma 24.2. *Let R be a ring with $1 \neq 0$, and let R^n be a free R-module of rank n with standard basis $e_i = (0, \ldots, 1, \ldots, 0)$. Then $Hom_R(R^n, R) \simeq R^n$, and it is generated by the dual basis e_j^*, defined by*

$$e_j^* : (\lambda_1, \ldots, \lambda_n) \mapsto \lambda_j.$$

Lemma 24.3. *Let R be a PID, N be a free R-module of rank n, and $M \leq N$ be a submodule. Then:*

1. *M is a free R-module of rank m, where $m \leq n$.*
2. *There exists a basis $z_1, \ldots, z_n$ of N and non-zero elements $d_1, \ldots, d_m \in R$ such that $d_i \mid d_{i+1}$ for all $1 \leq i < m$, and $d_1 z_1, \ldots, d_m z_m$ forms a basis for M.*

Proof. **Step 1: Construction of direct summands.**
We aim to find an element $z \in N$ ($z \neq 0$) and $d \neq 0$ in R such that $dz \in M$, Rz is a direct summand of N, and $R(dz)$ is a direct summand of M.

First, we claim that if $M \neq 0$, there exists $\psi \in \mathrm{Hom}_R(N, R)$ such that $\psi(M) \neq 0$. To show this, without loss of generality, assume $N = R^n$. Let $m = (a_1, \ldots, a_n) \in M$ be a non-zero element. Then $a_j \neq 0$ for some j. Using the dual basis functional, $e_j^*(m) = a_j \neq 0$, which implies $e_j^*(M) \neq 0$.

For any $\phi \in \mathrm{Hom}_R(N, R)$, the image $\phi(M)$ is an ideal of the PID R. Consider the set of ideals:

$$\Sigma = \{\phi(M) : \phi \in \mathrm{Hom}_R(N, R)\}.$$

Since $M \neq 0$, Σ is a non-empty set containing at least one non-zero ideal. Because R is a PID, it is Noetherian, so Σ possesses a maximal element with respect to inclusion. Choose $\psi \in \mathrm{Hom}_R(N, R)$ such that $\psi(M) = (d)$ is maximal in Σ. Choose $y \in M$ such that $\psi(y) = d$.

We claim that $d \mid \phi(y)$ for all $\phi \in \mathrm{Hom}_R(N, R)$. To show this, let $d' = \phi(y)$ and let $e = \gcd(d, d')$. Since R is a PID, we can write $e = \lambda d + \mu d'$ for some $\lambda, \mu \in R$. Then:

$$e = \lambda\psi(y) + \mu\phi(y) = (\lambda\psi + \mu\phi)(y) \in (\lambda\psi + \mu\phi)(M).$$

This implies that:

$$(d) \subseteq (e) \subseteq (\lambda\psi + \mu\phi)(M).$$

By the maximality of $(d) = \psi(M)$ in Σ, we must have $(d) = (e)$. Thus, d and e are associates, which means $d \mid d'$, and consequently $d \mid \phi(y)$.

Now, let $w_1, \ldots, w_n$ be a basis of N, and let $w_j^* \in \mathrm{Hom}_R(N, R)$ be the corresponding dual basis functionals. We can express $y = \sum_{j=1}^n \mu_j w_j$, meaning $w_j^*(y) = \mu_j$. By our previous

claim, $d \mid w_j^*(y)$, so $d \mid \mu_j$ for all j. Let:

$$z = \sum_{j=1}^{n} \left(\frac{\mu_j}{d}\right) w_j \in N.$$

Notice that $dz = y$. Furthermore, since $\psi(y) = d$, we have $\psi(dz) = d$. By linearity, $d\psi(z) = d$. Since R is an integral domain and $d \neq 0$, we conclude that $\psi(z) = 1$.

We now claim:

(a) $N = \ker(\psi) \oplus Rz$

(b) $M = (\ker(\psi) \cap M) \oplus Ry$

To show (a), note that if $x \in \ker(\psi) \cap Rz$, then $x = rz$ for some $r \in R$, and $0 = \psi(x) = r\psi(z) = r(1) = r$, so $x = 0$. For any $n \in N$, let $\lambda = \psi(n)$. We can write $n = (n - \lambda z) + \lambda z$. Observe that $\psi(n - \lambda z) = \lambda - \lambda\psi(z) = 0$, so $(n - \lambda z) \in \ker(\psi)$ and $\lambda z \in Rz$. Thus $N = \ker(\psi) \oplus Rz$. The proof of (b) is identical, substituting M for N and y for z.

Step 2: Proof of Part (1).
We proceed by induction on the rank of M (with N fixed). If the rank of M is 0, $M = 0$, which is trivially free. Assume the result holds for submodules of smaller rank. Using our construction, $M = Ry \oplus (\ker(\psi) \cap M)$. Because Ry is generated by a single non-torsion element, it has rank 1. Thus, the rank of $\ker(\psi) \cap M$ is strictly less than the rank of M. By the induction hypothesis, $\ker(\psi) \cap M$ is a free R-module. Adding y to a basis of $\ker(\psi) \cap M$ yields a free basis for M. Thus M is free of rank $m \leq n$.

Step 3: Proof of Part (2).
We proceed by induction on the rank n of N, proving the statement for all submodules $M \leq N$ simultaneously. If $n = 0$, it is trivial. Suppose it holds for free modules of rank $n - 1$. From Step 1, $N = \ker(\psi) \oplus Rz$. Since N has rank n and Rz has rank 1, $\ker(\psi)$ is a free R-module of rank $n - 1$. By the induction hypothesis applied to the submodule $\ker(\psi) \cap M \leq \ker(\psi)$, there exists a basis $z_2, \ldots, z_n$ of $\ker(\psi)$ and elements $d_2, \ldots, d_m \in R$ with $d_2 \mid d_3 \mid \cdots \mid d_m$ such that $d_2 z_2, \ldots, d_m z_m$ is a basis for $\ker(\psi) \cap M$.

Let $z_1 = z$ and $d_1 = d$. Then $z_1, z_2, \ldots, z_n$ is a basis for N, and $d_1 z_1, d_2 z_2, \ldots, d_m z_m$ is a basis for M.

It remains only to show that $d_1 \mid d_2$. Define a homomorphism $f : N \to R$ by mapping $z_1 \mapsto 1$, $z_2 \mapsto 1$, and $z_i \mapsto 0$ for $i > 2$. We evaluate $f(M)$: $f(d_1 z_1) = d_1$ and $f(d_2 z_2) = d_2$, so the ideal $f(M)$ contains both d_1 and d_2, meaning $f(M) = (d_1, d_2)$. Because $(d_1) = \psi(M)$ is maximal among such ideals, and $(d_1) \subseteq (d_1, d_2) = f(M)$, we must have $(d_1) = (d_1, d_2)$. This implies $d_2 \in (d_1)$, so $d_1 \mid d_2$. This completes the proof. $\square$

25 March 20

Theorem 25.1. *Let F be a finitely genereated module over PID R say $F = Rf_1 + \cdots + Rf_t$, $f_i \in F$. Define $\pi : R^t \to F$ by $\pi : (\lambda_1, ..., \lambda_t) \mapsto \sum_{i=1}^t \lambda_i t_i$. π is a surjective homomorphism so*

$$F \simeq \frac{N}{\ker(\pi)}.$$

We have N, M having rank t and s.

Theorem 25.2. *Let R be a PID, M be a finitely generated R-module. Then M is isomorphic to*

$$\frac{R}{(c_1)} \oplus \cdots \oplus \frac{R}{(c_n)} \oplus R^t$$

where $t \geq 0$, $c_i \neq 0$, $c_i \mid c_{i+1}$ for all $1 \leq i < n$, and the elements c_i are non-units.

Lemma 25.3. *By previous result, we get a basis $n_1, ..., n_t$ for N and nonzero constants $r_1, ..., r_s \in R$ such that $r_i | r_{i+1}$ for $1 \leq i < s$ and $r_1 n_1, ..., r_s n_s$ is a basis for M.*

Exercise 25.4.

$$\frac{N}{M} \simeq \frac{R}{(r_1)} \oplus \cdots \oplus \frac{R}{(r_s)} \oplus R^{t-s}.$$

Lemma 25.5. *If r nonzero nonunit, $r = u\pi_1^{k_1} \cdots \pi_n^{k_n}$ for u unit, π non associate and irreducibles.*

Theorem 25.6 (Chinese Remainder Theorem)**.** *If $(r) = (\pi_1)^{k_1} \cdots (\pi_n)^{k_n}$ then*

$$\frac{R}{(r)} \simeq \frac{R}{(\pi_1^{k_1})} \oplus \cdots \oplus \frac{R}{(\pi_n^{k_n})}.$$

25.1 Adjoint Functors

Definition 25.7. *Let $\mathcal{U}$ be "forgetful" functor from group to set.*

$$\mathcal{U}(G) = \text{underlying set of } G.$$

If $\phi : G \to H$, $\mathcal{U}(\phi)$ is ϕ viewed as a function.

Lemma 25.8. *We have for any group G and any set X, there is a bijection between $Hom_{Gr}(Fr(X), G)$ and $Hom_{Set}(X, \mathcal{U}(G))$*

Definition 25.9 (Natural Transformation)**.** *Let $F_1, F_2 : \mathcal{C} \to \mathcal{D}$. A natural transformation from $F_1 \to F_2$ is a family $v = (v_c)$, for c object of $\mathcal{C}$, $F_1 c \xrightarrow{v_c} F_2 c$ in $\mathcal{D}$ such that for all $c \xrightarrow{\alpha} c'$*

$$\begin{tikzcd} F_1 c \arrow[r, "v_c"] \arrow[d, "F_1\alpha"'] & F_2 c \arrow[d, "F_2\alpha"] \\ F_1 c' \arrow[r, "v_{c'}"'] & F_2 c' \end{tikzcd}$$

commutes.

Definition 25.10 (Opposite Category)**.** *$\mathcal{C}$ category then $\mathcal{C}^{op}$ is the category with the same objects but with the arrows reversed.*